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Experiment 2: Candidate Elimination Algorithm

To implement **Candidate-Elimination algorithm** using training data stored in a **CSV file** and display:

Specific Hypothesis S
General Hypothesis G

Python Program:

```
import pandas as pd

from google.colab import files

uploaded = files.upload()

filename = list(uploaded.keys())[0]

data = pd.read_csv(filename)

concepts = data.iloc[:, :-1].values

target = data.iloc[:, -1].values

specific_h = concepts[0].copy()

general_h = [["?" for i in range(len(specific_h))] for i in range(len(specific_h))]

print("Initial Specific Hypothesis:")

print(specific_h)

print("\nInitial General Hypothesis:")

for row in general_h:

    print(row)

for i, h in enumerate(concepts):

    if target[i] == "Yes":

        for x in range(len(specific_h)):
```

```
            if h[x] != specific_h[x]:
                specific_h[x] = '?'
                general_h[x][x] = '?'
    if target[i] == "No":
        for x in range(len(specific_h)):
            if h[x] != specific_h[x]:
                general_h[x][x] = specific_h[x]
            else:
                general_h[x][x] = '?'
print("\nFinal Specific Hypothesis:")
print(specific_h)
```

Output

```
Initial Specific Hypothesis:
['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same' 'Summer' 'Outdoor'
 'Morning']

Initial General Hypothesis:
['?', '?', '?', '?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?', '?', '?', '?']

Final Specific Hypothesis:
['?' '?' '?' '?' '?' 'Same' '?' 'Outdoor' '?']

Final General Hypothesis:
['?', '?', '?', '?', '?', 'Same', '?', '?', '?']
['?', '?', '?', '?', '?', '?', '?', 'Outdoor', '?']
```